

WEDE Project Proposal

WEDE5020
Web Development

Website Project Proposal

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Project Proposal Content

---IIE website proposal---

In my first semester of university at RCI, I was introduced to the IIE Harvard Anglia referencing system. Which was a completely new and refined referencing system adapted for IIE tertiary education.

Back in High school, I would often make use of reference generators found online such as MyBib (<https://www.mybib.com/tools/harvard-referencing-generator>) which would save me a bunch of time when trying to reference websites, books, or blogs for my school projects. But now, inside RCI, none of the students have access to a resource like this. Having them spend hours of extra time making sure to format their references in the IIE style, which could instead be used to refine their project or assignment even further.

This is why I have come up with the idea of creating an online reference generator specifically for the IIE referencing system, allowing students to worry about the work that really matters, and not stress about references and guarantee the small amount of marks awarded for correct referencing.

1. Organisational Overview

Target Organisation

Name: IIE Rosebank International

Brief History:

IIE Rosebank International is one of South Africa's leading private tertiary institutions, operating as an educational branch of the Independent Institute of Education (IIE) which is a subsidiary of the largest accredited higher education provider, ADvTECH Group.

Rosebank College was founded and established originally in 1948, delivering accessible, employment-focused skill training. It was later integrated into the IIE in 2005 cementing its academic governance, receiving accreditation under the Department of Higher Education and having a quality assurance standard recognised internationally by the British Accreditation Council.

Over the next 2 decades, Rosebank college will have expanded from a localised training centre to a fully-fledged tertiary educational institution. Today it operates 9 campuses across South Africa – including Braamfontein, Cape Town, Pretoria, Durban, Bloemfontein, and Polokwane – serving over 30,000 students through a full-time, hybrid, and distance learning models.

2. Website Goals and Objectives

Goals

- This website will effectively increase student productivity and willingness to complete projects and assignments. As referencing is a tedious, repetitive task, with many students relying on online reference generators in High School. I believe a dedicated website for generating references adapted for the IIE Harvard Anglia referencing system will benefit all students attending an IIE institute.
- My goal is to create a fully functioning website with an interactable UX and a well designed UI. Designed for students that will utilize it to save time when referencing inside a project. Allowing them to focus on the more "mark-heavy" detailed answers.

Key Performance Indicators (KPI's)

Audience	Audience's Need	Web Dev Need	Content/ Feature	KPI
Website User/ Audience	Select Reference Tool	User Login	A webpage that generates references based on inputted information	%of visits cf. users who utilise webpage functions
	Input Reference Material [i.e. Website Link, Book Name & Author]	Any unknown referencing formats that need to be implemented	↓↑	%of user inputs and generated text
	Obtain Formatted Reference [in-text and citation]		Copiable text and IIE formatted reference	
	Feedback and Help Centre	User feedback on generated references and site UX/UI	Separate webpage for user feedback and FAQ	ratings and %of issues compared praises given

3. Proposed Website Features and Functionality

Landing Page

Header

Navigation Pane

Website Label, Dev info, RI info, Products, FAQ

users can click these text buttons which will take them too the desired page. Will maybe implement a side/drop-down menu instead of a navigation bar to make the website more modern/codebase inspired look

Main

Reference Type Selector

drop-down menu, searchable

users will be able to search for the reference type they would like to generate. Had the thought of adding an "auto-detection" feature, however I think it would be more tedious and error prone. So allowing the user to select the specific type seems more applicable

Input field (dependent on selected type)

Books (one||multiple authors), website, pdf doc, video, etc.

a separate input field skeleton will appear depending on the selected reference type. For example if the user selects "website", the website will display a textbox in which they can input the URL, or if it is a BLOG, they will enter both the URL and the author of the page.

Buttons

- Confirmation
- Generate
- Refresh
- Clear

Footer

- Contact Us
- Copyright
- extra information
- hyperlinks

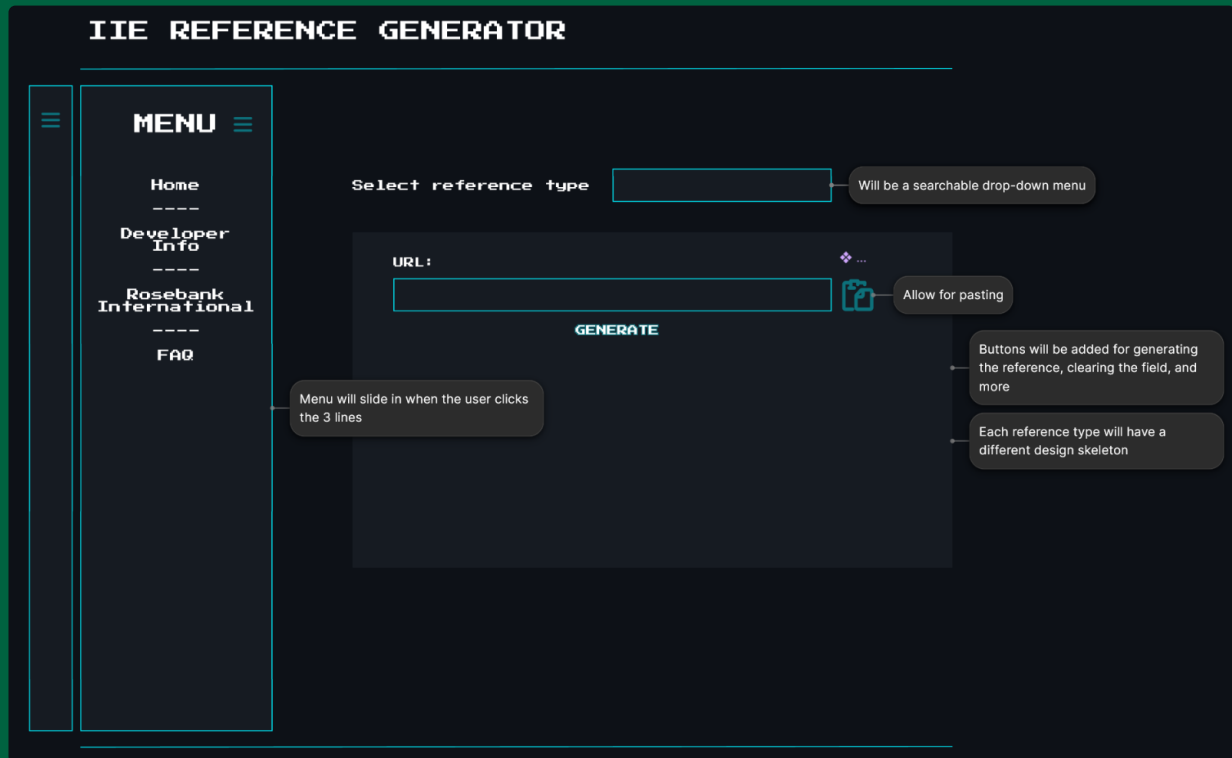
4. Design and User Experience

Design & UI/UX Design System

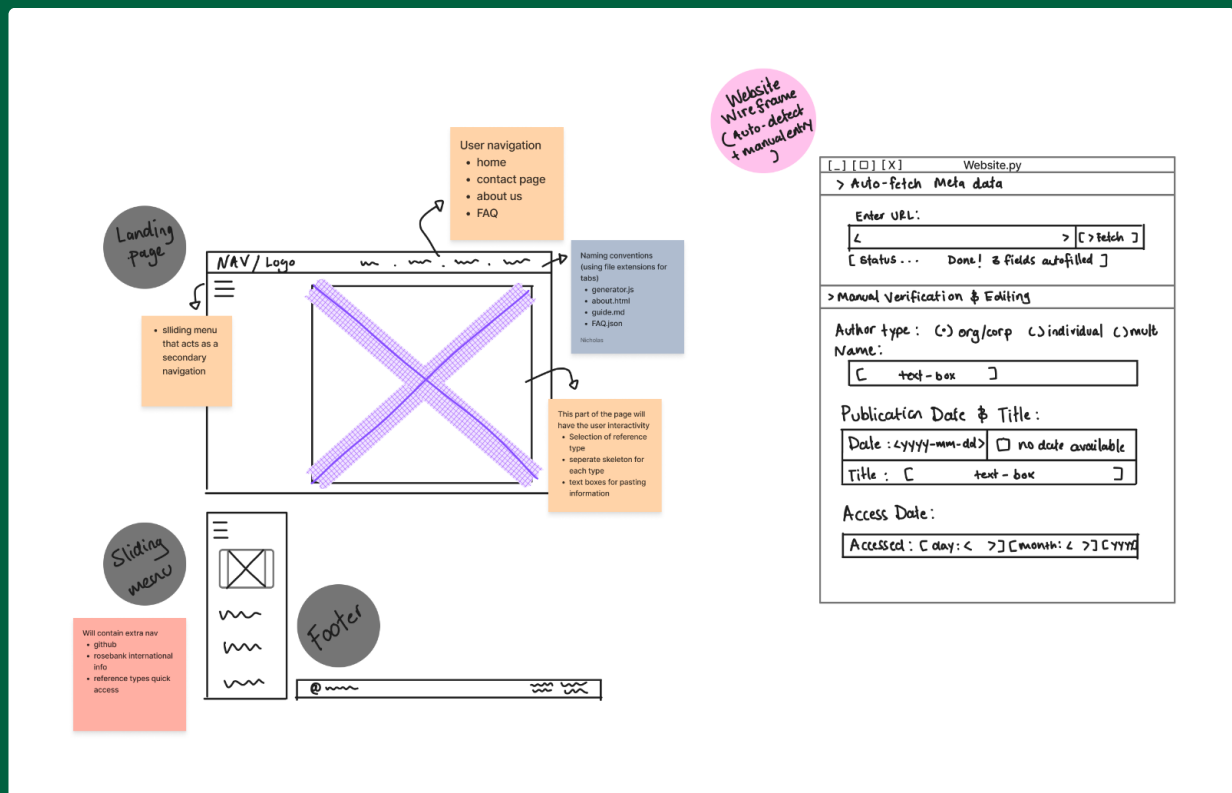
Design Category	Recommendation	Details & Description
Primary Aesthetic Concept	Retro IDE / Pixel Codebase	Blends a retro operating system layout with a modern code editor and subtle pixel art details. Think windowed frames, tabbed code views, and line numbers.
Color Palette 1 (Cyberpunk Syntax)	• Background: #0D1117 (Dark Navy) • UI Frame: #161B22 (Charcoal) • Primary Accent: #00F0FF (Cyan) • Secondary Accent: #FF0055 (Neon Pink) • Text: #E6EDF3 (Soft White)	High contrast, modern developer aesthetic. Great for highlighting output strings (like generated in-text vs. bibliography citations).
Typography (Headers / Accents)	Pixel Display Fonts: • Press Start 2P • Silkscreen • VT323	Best used sparingly for window headers, title bars, action buttons, and decorative badges.

Typography (Body / Input / Output)	Monospace Fonts: • JetBrains Mono • Fira Code • Roboto Mono	Essential for inputs, forms, and final formatted citations. Highly legible for academic text while keeping the coding motif intact.
Window & Frame Style	Pixelated Windows with IDE Tabs	Framed container boxes with window control buttons (<code>[_] [□] [X]</code>) or tabbed panels (e.g., <code>Book.exe</code> , <code>Journal.js</code> , <code>Website.py</code>) for switching source types.
Buttons & Controls	Pixel/Retro Tactile Buttons	Flat or 3D-bevelled pixelated buttons that give a satisfying "press down" state on click, complete with classic retro hover effects.
Feedback & Notifications	Terminal Status Bar	A bottom status bar (like a terminal output pane) showing messages like <code>> Citation compiled successfully!</code> or <code>> Error: Missing Publication Year.</code>

Low-fidelity Wireframe



created using figma*



Technical Requirements

Programming Languages, Frameworks & Libraries

Technical Layer	Technology	Purpose & Usage
Markup Language	HTML5	Provides the semantic structure of the website, including input forms for reference details, source tabs, output display containers, and modal dialogs.
Styling & Design	CSS	Handles custom layouts (Flexbox/CSS Grid), color palettes, animations, and retro UI styling (pixel borders, custom retro fonts, dark mode code theme).
CSS Framework / Utility (<i>Extra research</i>)	NES.css or 98.css (<i>Optional</i>)	Lightweight, pure CSS frameworks for instant retro pixelated component styling (retro buttons, inputs, dialog boxes, and window containers).
Client-Side Scripting	JavaScript	Drives core application logic: dynamically parsing user input, applying citation string formatting algorithms (IIE Harvard Anglia rules), Document Object Model (DOM) manipulation, and single-click copy functionality.

Typography	Google Fonts API (optional)	Imports retro pixel fonts (e.g., <i>Press Start 2P</i> , <i>Silkscreen</i>) for UI elements and clean monospace fonts (e.g., <i>JetBrains Mono</i> , <i>Fira Code</i>) for input fields and generated output strings.
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Hosting

Providers:

→ GitHub, Vercel, Netlify, Google Cloud Web Hosting

I will have low requirements as the entire site is made up of front-end static files which are < 5MB. Perfect for free tier 100GB/month limits. All the calculation happens right inside the users browser.

Environment:

→ Static web hosting

Webpages are served to users as they are saved on the server (runs files directly on the browser) – strictly for HTML, css, and Javascript as I am not relying on any back-end services like MySQL for database management and server side languages like python or Node.js

Security

→ SSL/TLS security provided with HTTPS

Enabled via host-provided automatic SSL certificates to ensure a secure connection

Deployment Workflow

Continuous Deployment (CD) connected directly to the project's GitHub repository for automatic updates upon git commits.

Domain Name Requirements

I will utilise GitHub hosting as it is free and opensource.
e.g. `username.github.io/reference-generator`
Or a custom `.com` / `.ac.za` domain using Google web hosting or Vercel/Netlify

Timelines and Milestones

Phase & Dates	Phase Name	Key Milestones & Tasks	Key Deliverables
Phase 1 03 Aug – 09 Aug	Planning & Wireframing	• Define core project requirements and IIE Harvard Anglia referencing rules. • Set up local development environment (VS Code, Git repository). • Draft visual mockups and choose color palette/fonts for the retro pixel aesthetic.	• Approved Project Proposal document. • Initialized Git repo with basic folder structure (index.html, style.css, app.js).
Phase 2 10 Aug – 16 Aug	HTML Structure & CSS Styling	• Build semantic HTML layout (header, source type selection tabs, input form fields, output container). • Apply retro pixel aesthetic with custom CSS (window frames, retro borders, fonts, and responsive layout). • Create static preview of 12 formatted citation boxes.	• Implemented basic page layout. • Styled retro UI components and responsive window containers.

Phase 3 17 Aug – 23 Aug	JavaScript Logic Implementation	• Write JavaScript functions to capture user input data from form fields. • Implement IIE Harvard Anglia formatting rules dynamically (handling italics, author lists, dates, and URLs). • Program source type switching logic (Book, Journal, Website forms). • Add single-click "Copy to Clipboard" functionality.	• Fully functional reference generation algorithm in JavaScript. • Working DOM manipulation for dynamic input/output rendering.
Phase 4 24 Aug – 30 Aug	Testing, Polish & Deployment	• Perform cross-browser testing and responsive screen checks. • Validate IIE Harvard Anglia formatting against official guide examples. • Fix visual bugs, fine-tune pixel aesthetic spacing, and add subtle animations. • Deploy static site live to GitHub Pages / Vercel.	• Final live website URL. • Clean, commented codebase ready for final submission.

Budget

Estimated Cost Breakdown

- **Domain Registration (Annual Recurring)**

- **Estimated Cost:** ~R150 – R250 / year
- **Details:** Registration for a custom domain (such as `.co.za` or `.com`) via registrars like Namecheap or local providers.

- **Web Hosting (Free Tier)**

- **Estimated Cost:** R0.00
- **Details:** Static site hosting provided via GitHub Pages, Vercel, or Netlify.

- **Development & Design**

- **Estimated Cost:** ~R3,000 – R5,000
- **Details:** Estimated front-end UI/UX design and JavaScript logic implementation (~30 hours of development time based on standard freelance/student entry rates).

- **Maintenance & Security**

- **Estimated Cost:** R0.00
- **Details:** Automatic SSL certificate renewal and automated deployments via GitHub Actions integrated directly into hosting.

- **Total Estimated Project Cost**

- **First-Year Implementation:** ~R3,150 – R5,250

obviously based in fantasy as this website will be locally run throughout the semester, however, it is good practice as I might publish it as an official website if it gains recognition.

Folder Hierarchy

```
2  ---📁 iie-harvard-anglia-reference-generator/
3    |
4    ---> index.html
5    ---> about.html
6    ---> contact.html
7    ---> README.md
8    |
9    📁 css-assets/
10   |
11   --->
12   |
13   📁 js-assets/
14   |
15   --->
16   |
17   📁 images/
18   |
19   --->
20   |
21   📁 documents/
22   |
23   ---> Project_documentation.pdf
```

Reference Index

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